

In this section, an understanding of non-communicable diseases will be illustrated by reference to cancer, smoking-induced lung diseases and asthma.

Learners will use their knowledge of cell division from Module 3.1 to further their understanding of the cellular and genetic basis of cancer. The computerised technological devices and tests that enable early detection and diagnosis of cancer will also be considered, as well as the use of genetic screening to assess potential risks.

Consideration will be given to the analysis of epidemiological data when evaluating the causes of non-infectious diseases, and learners will consider the importance of clinical trials when developing new medicinal treatments. The role of government health agencies in providing national guidelines for treatments is also explored.

In this section, learners will consider the biological and ethical implications of medical and scientific advancements and evaluate the associated risks and benefits to society.

Learners are expected to apply knowledge, understanding and other skills gained in this section to new situations and/or to solve related problems.

3.3.1 The cellular basis of cancer and treatment

Learning outcomes	Additional guidance
<i>Learners should be able to demonstrate and apply their knowledge and understanding of:</i>	
(a) the factors that may increase the risk of developing non-communicable diseases	Factors to include heredity, ageing, types of radiation, carcinogen, viruses and air pollution and diseases to include cancers and asthma AND to include an evaluation of epidemiological and other evidence to identify correlations.
(b) the evidence for causal relationships between certain factors and lung disease	Factors to include heredity, ageing, types of radiation, carcinogen, viruses and air pollution AND to include an evaluation of epidemiological and other evidence to identify correlations.
(c) the cellular basis of cancer	To include an outline of cell cycle control and the changes in control which lead to the formation of tumours and metastases.
(d) how mutations to proto-oncogenes can lead to cancer	To include Ras and Myc proto-oncogenes.
(e) how mutations to tumour suppressor genes can lead to cancer	To include the p53 gene.

- (f) the evaluation of epidemiological evidence linking potential risk factors with particular forms of cancer
- To include smoking and lung cancer, diet and bowel cancer, BRCA1 gene mutations and breast cancer.
M1.3, M1.5, M1.7, M3.1
HSW5, HSW6
- (g) the methods used to detect cancers
- To include references to MRI, X-rays, mammography, CT scans, ultrasound, PET scans, biopsies and blood tests.
- (h) the ethical and economic considerations when screening and conducting genetic tests for cancer
- To include evaluation of screening for particular cancers e.g. the potential harm, accuracy and cost of the screening procedure
AND
discussion of the ethics of genetic tests e.g. for BRCA and HNPCC genes.
HSW9, HSW10
- (i) the methods used to treat patients with cancer.
- To include surgery, chemotherapy, radiotherapy, immunotherapy (monoclonal antibodies), complementary therapies and hormone-related treatment.